

EDA PROJECT
HR Employee Attrition Analysis using Python
CodeAlpha Data Analytics Internship

1. Executive Summary

- Analyzed the IBM HR Employee Attrition dataset using Python.
 - Performed data cleaning and exploratory data analysis (EDA).
 - Identified the major factors contributing to employee attrition.
 - Used statistical analysis and visualizations to generate business insights.
 - Proposed recommendations to improve employee retention.
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2. Introduction

- Employee attrition is a significant challenge for organizations.
 - High attrition leads to increased recruitment costs and productivity loss.
 - This project analyzes employee data to identify the primary reasons for employee turnover.
 - The analysis helps HR departments make data-driven decisions.
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3. Objectives

- Understand employee demographics.
 - Perform data cleaning and preprocessing.
 - Explore relationships between employee attributes and attrition.
 - Identify high-risk employee groups.
 - Provide actionable business recommendations.
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4. Dataset Overview

- Dataset Name: IBM HR Employee Attrition Dataset
 - Total Records: **1470**
 - Total Features: **35**
 - Target Variable: **Attrition**
 - Tools Used:
 - Python
 - Google Colab
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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5. Data Cleaning

- Loaded the dataset.
 - Checked dataset dimensions.
 - Verified data types.
 - Examined missing values.
 - Checked duplicate records.
 - Generated descriptive statistics.
 - Created Age Group categories for analysis.
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6. Exploratory Data Analysis (EDA)

Question 1

Which department has the highest employee attrition?

Observation

- Research & Development has the highest number of employee exits (133).
- Sales follows with 92 exits.
- Human Resources has 12 exits.

Business Insight

- R&D has the largest workforce, leading to the highest number of resignations.
 - HR should investigate workload, engagement, and career growth opportunities in R&D.
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Question 2

Which department has the highest attrition rate (%)?

Observation

- Sales: **20.6%**
- Human Resources: **19.0%**
- Research & Development: **13.8%**

Business Insight

- Although R&D has more resignations, employees in Sales are proportionally more likely to leave.
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Question 3

Which age group has the highest attrition?

Observation

- Employees aged **26–35 years** show the highest attrition.

Business Insight

- Younger employees are more likely to switch jobs for better career opportunities.
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Question 4

Does overtime affect employee attrition?

Observation

- Employees working overtime experience significantly higher attrition.

Business Insight

- Excessive workload negatively impacts employee retention.
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Question 5

Which job role has the highest attrition?

Observation

- Laboratory Technicians and Sales Executives have the highest employee exits.

Business Insight

- These roles require focused retention and engagement strategies.
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Question 6

Does monthly income affect attrition?

Observation

- Employees leaving the organization generally earn lower monthly incomes.

Business Insight

- Competitive compensation can improve employee retention.
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Question 7

Does job satisfaction affect attrition?

Observation

- Employees with lower job satisfaction show higher attrition.

Business Insight

- Improving workplace satisfaction can reduce employee turnover.
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Question 8

Which marital status has the highest attrition?

Observation

- Single employees have the highest attrition.

Business Insight

- Early-career employees may require better career development opportunities.
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Question 9

Does business travel affect attrition?

Observation

- Travel_Rarely has the highest number of employee exits.

Attrition Rate Analysis

Observation

- Travel_Frequently: **24.9%**
- Travel_Rarely: **15.0%**
- Non-Travel: **8.0%**

Business Insight

- Employees travelling frequently are more likely to leave the organization.
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Question 10

Which education field has the highest attrition?

Observation

- Life Sciences contributes the highest number of employee exits.

Business Insight

- HR should investigate retention strategies for this employee group
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Question 11

Does gender affect attrition?

Observation

- Male employees record more attrition cases than female employees.

Business Insight

- Additional analysis is required to determine whether this is due to workforce size or other factors.
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Question 12

Does total working experience affect attrition?

Observation

- Employees with fewer years of work experience are more likely to leave.

Business Insight

- Early-career employees benefit from mentoring and structured career development.
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7. Key Findings

- Research & Development has the highest number of employee exits.
 - Sales has the highest attrition rate.
 - Employees aged 26–35 years are more likely to leave.
 - Overtime significantly increases attrition.
 - Laboratory Technicians and Sales Executives have higher turnover.
 - Lower income is associated with higher attrition.
 - Lower job satisfaction contributes to resignations.
 - Single employees have higher attrition.
 - Frequent business travel increases employee turnover.
 - Employees with fewer years of experience are more likely to resign.
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8. Business Recommendations

- Develop targeted retention programs for the Sales department.
 - Improve employee engagement within the R&D department.
 - Reduce excessive overtime and promote work-life balance.
 - Review employee compensation policies.
 - Enhance career development and learning opportunities.
 - Conduct regular employee satisfaction surveys.
 - Strengthen mentoring programs for early-career employees.
 - Introduce flexible work arrangements where possible.
 - Continuously monitor attrition using HR analytics dashboards.
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9. Conclusion

- Employee attrition is influenced by multiple factors, including department, age, overtime, income, job satisfaction, business travel, and work experience.
 - A data-driven HR strategy can help reduce employee turnover.
 - The insights generated from this analysis support better workforce planning and employee retention initiatives.
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10. Future Scope

- Build a Machine Learning model to predict employee attrition.
- Develop an interactive HR dashboard using Power BI or Tableau.
- Perform employee segmentation using clustering techniques.
- Apply advanced predictive analytics to identify employees at risk of leaving.
- Monitor HR metrics through real-time dashboards.



THANK YOU